

Module 6: ES6+ Features in JavaScript

ES6 (ECMAScript 2015) introduced many powerful features that make JavaScript easier, cleaner, and more modern.

In this module, we will cover:

1. let
2. const
3. Template Literals
4. Modules
5. Rest Operator (...)
6. Spread Operator (...)

1. let Keyword

let is used to declare variables that can be reassigned.

Example

```
let name = "Vicky";
```

```
console.log(name);
```

```
name = "Rahul";
```

```
console.log(name);
```

Output

Vicky

Rahul

Block Scope of let

```
let age = 20;
```

```
if(age >= 18) {  
  let message = "Eligible";  
  console.log(message);  
}
```

```
// console.log(message); ❌ Error
```

Explanation

Variables declared with let are only accessible inside the block {} where they are created.

2. const Keyword

const creates a constant variable.

Its value cannot be reassigned.

Example

```
const PI = 3.14159;
```

```
console.log(PI);
```

```
// PI = 3.14; ❌ Error
```

Object with const

```
const student = {  
  name: "Vicky"  
};  
  
student.name = "Rahul";  
  
console.log(student.name);
```

Output

Rahul

Important

const prevents reassignment of the variable, not modification of object properties.

var vs let vs const

Feature	var	let	const
Reassign	☐	☐	☐
Redeclare	☐	☐	☐
Block Scope	☐	☐	☐

Example

```
var a = 10;  
var a = 20; // Allowed
```

```
let b = 10;  
// let b = 20; ❌ Error
```

```
const c = 30;  
// c = 40; ❌ Error
```

3. Template Literals

Template Literals use backticks ` instead of quotes.

They allow:

- Variable interpolation
 - Multi-line strings
-

Old Method

```
let name = "Vicky";  
  
console.log("Hello " + name);
```

ES6 Method

```
let name = "Vicky";  
  
console.log(`Hello ${name}`);
```

Output

Hello Vicky

Multiple Variables

```
let name = "Vicky";  
let age = 20;
```

```
console.log(  
  `My name is ${name} and I am ${age} years old.`  
);
```

Output

My name is Vicky and I am 20 years old.

Multi-Line Strings

```
let msg = `  
Welcome  
to  
JavaScript  
`;
```

```
console.log(msg);
```

Output

Welcome
to
JavaScript

Practical Example: Student Result

```
let student = "Amit";  
let marks = 85;  
  
console.log(  
  `${student} scored ${marks}% in the examination.`  
);
```

Output

Amit scored 85% in the examination.

4. Modules

Modules help split JavaScript into multiple files.

Benefits:

- Better organization
 - Reusability
 - Easier maintenance
-

math.js

```
export function add(a, b) {  
  return a + b;  
}  
  
export function multiply(a, b) {  
  return a * b;  
}
```

app.js

```
import { add, multiply } from "./math.js";
```

```
console.log(add(10, 20));
```

```
console.log(multiply(5, 4));
```

Output

30

20

Default Export

message.js

```
export default function greet() {  
  console.log("Welcome");  
}
```

app.js

```
import greet from "./message.js";
```

```
greet();
```

Output

Welcome

Practical Example: GST Module

gst.js

```
export function calculateGST(amount) {  
  return amount * 0.18;  
}
```

app.js

```
import { calculateGST }  
from "./gst.js";  
  
console.log(calculateGST(1000));
```

Output

180

5. Rest Operator (...)

Rest Operator collects multiple values into a single array.

Example

```
function total(...numbers) {  
  
  console.log(numbers);  
}  
  
total(10,20,30,40);
```

Output

[10,20,30,40]

Sum Using Rest Operator

```
function sum(...numbers) {  
  
    return numbers.reduce(  
        (total,num) => total + num,  
        0  
    );  
}
```

```
console.log(  
    sum(10,20,30,40)  
);
```

Output

100

Practical Example: Shopping Cart

```
function cartTotal(...prices) {  
  
    return prices.reduce(  
        (sum, price) => sum + price,  
        0  
    );  
}
```

```
console.log(
```

```
cartTotal(500,1000,2000)
);
```

Output

3500

6. Spread Operator (...)

Spread Operator expands arrays or objects.

Array Spread

```
let arr1 = [1,2,3];

let arr2 = [...arr1,4,5];

console.log(arr2);
```

Output

[1,2,3,4,5]

Merge Arrays

```
let boys =
["Amit","Rahul"];

let girls =
["Pooja","Neha"];
```

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```
let students =  
[...boys,...girls];
```

```
console.log(students);
```

Output

```
["Amit","Rahul","Pooja","Neha"]
```

Copy Array

```
let numbers = [1,2,3];
```

```
let copy = [...numbers];
```

```
console.log(copy);
```

Output

```
[1,2,3]
```

Object Spread

```
let student = {  
  name:"Vicky"  
};
```

```
let details = {  
  ...student,  
  age:20,  
  city:"Jaipur"  
};
```

```
console.log(details);
```

Output

```
{  
  name:"Vicky",  
  age:20,  
  city:"Jaipur"  
}
```

Rest vs Spread

Rest	Spread
Collects values	Expands values
Used in parameters	Used in arrays/objects
Creates array	Breaks array

Rest Example

```
function show(...items) {  
  console.log(items);  
}
```

```
show(1,2,3,4);
```

Output

```
[1,2,3,4]
```

Spread Example

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```
let nums = [1,2,3];
```

```
console.log(...nums);
```

Output

```
1 2 3
```

Real-World Example: Online Shopping Cart

```
const products = [500,1000,1500];
```

```
const extraProducts = [2000,2500];
```

```
const allProducts = [  
  ...products,  
  ...extraProducts  
];
```

```
const total =  
allProducts.reduce(  
  (sum, price) => sum + price,  
  0  
);
```

```
console.log(  
  `Total Bill = ₹${total}`  
);
```

Output

```
Total Bill = ₹7500
```

Interview Questions

Q1. Difference between let and const?

Answer:

- let can be reassigned.
 - const cannot be reassigned.
-

Q2. Why use Template Literals?

Answer:

To embed variables and create multi-line strings easily.

Q3. What is Rest Operator?

Answer:

It collects multiple arguments into an array.

Q4. What is Spread Operator?

Answer:

It expands arrays or objects into individual elements.

Q5. Why use Modules?

Answer:

To organize code into reusable files and avoid global variable conflicts.

Comprehensive Practice Program

```
const student = "Vicky";

function totalMarks(...marks) {

    return marks.reduce(
        (sum, mark) => sum + mark,
        0
    );
}

let result =
totalMarks(80,90,85);

console.log(
`${student} scored ${result} marks`
);
```

Output

Vicky scored 255 marks

Summary

Topic	Purpose
let	Block-scoped variable

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Topic	Purpose
const	Constant variable
Template Literals	Easy string formatting
Modules	Split code into files
Rest Operator	Collect values
Spread Operator	Expand values

These ES6+ features are used heavily in **React, Node.js, Vue.js, Angular, APIs, modern web applications, and enterprise JavaScript development.**